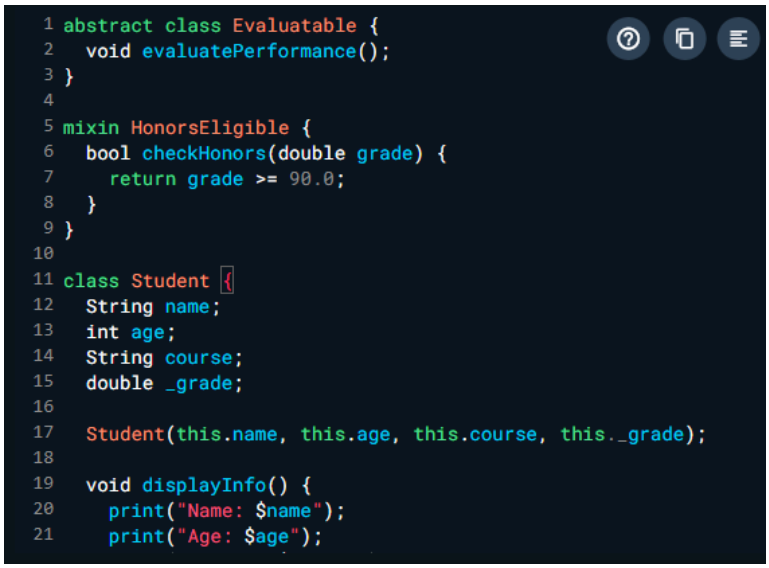
National Teachers College

629 J Nepomuceno St, Quiapo, Manila, 1001 Metro Manila

Catherine Jillian M. Inocencio 3.1 BSIT

Week 11 Your Turn: Add Async Behavior to Your OOP App

Code:



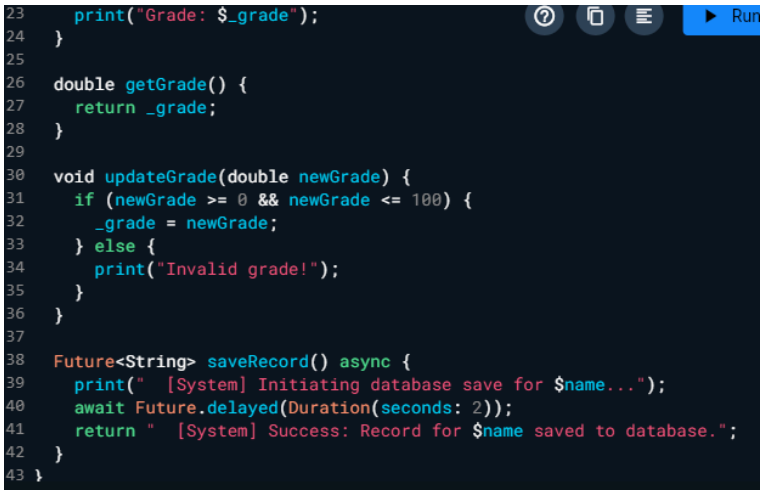
```dart
abstract class Evaluatable {
  void evaluatePerformance();
}

mixin HonorsEligible {
  bool checkHonors(double grade) {
    return grade >= 90.0;
  }
}

class Student {
  String name;
  int age;
  String course;
  double _grade;

  Student(this.name, this.age, this.course, this._grade);

  void displayInfo() {
    print("Name: $name");
    print("Age: $age");
```



```dart
    print("Grade: $_grade");
  }

  double getGrade() {
    return _grade;
  }

  void updateGrade(double newGrade) {
    if (newGrade >= 0 && newGrade <= 100) {
      _grade = newGrade;
    } else {
      print("Invalid grade!");
    }
  }

  Future<String> saveRecord() async {
    print("  [System] Initiating database save for $name...");
    await Future.delayed(Duration(seconds: 2));
    return "  [System] Success: Record for $name saved to database.";
  }
}
```

National Teachers College

629 J Nepomuceno St, Quiapo, Manila, 1001 Metro Manila

```
class UndergraduateStudent extends Student implements Evaluatable {
  String yearLevel;

  UndergraduateStudent(
    String name,
    int age,
    String course,
    double grade,
    this.yearLevel,
  ) : super(name, age, course, grade);

  @override
  void displayInfo() {
    super.displayInfo();
    print("Year Level: $yearLevel");
    print("Status: Undergraduate");
    print("--------------------");
  }

  @override
  void evaluatePerformance() {
    if (getGrade() >= 75.0) {
      print("$name is passing their undergraduate modules.");
    } else {
      print("$name needs academic support.");
    }
  }
}

class GraduateStudent extends Student implements Evaluatable {
  String thesisTopic;
```

National Teachers College

629 J Nepomuceno St, Quiapo, Manila, 1001 Metro Manila

```
77  GraduateStudent(
78    String name,
79    int age,
80    String course,
81    double grade,
82    this.thesisTopic,
83  ) : super(name, age, course, grade);
84
85  @override
86  void displayInfo() {
87    super.displayInfo();
88    print("Thesis Topic: $thesisTopic");
89    print("Status: Graduate");
90    print("--------------------");
91  }
92
93  @override
94  void evaluatePerformance() {
95    if (getGrade() >= 85.0) {
96      print("$name is meeting graduate research standards.");
97    } else {
```

```
98      print("$name's research progress is under review.");
99    }
00  }
01 }
02
03 class AcademicAdvisor with HonorsEligible {
04   String name;
05   String department;
06
07   AcademicAdvisor(this.name, this.department);
08
09   void reviewStudent(Student student) {
10     bool isHonors = checkHonors(student.getGrade());
11     print("Advisor $name ($department):");
12     print(
13       "${student.name} Honors Status: ${isHonors ? 'Eligible' : 'Not
```

National Teachers College

629 J Nepomuceno St, Quiapo, Manila, 1001 Metro Manila

```
  Eligible'}",
114     );
115     print("--------------------");
116   }
117 }
118
119 void main() async {
120   UndergraduateStudent alice = UndergraduateStudent(
121     "Alice",
122     20,
123     "Information Technology",
124     90.5,
125     "3rd Year",
126   );
127
128   GraduateStudent bob = GraduateStudent(
```

```
    "Bob",
    25,
    "Computer Science",
    85.0,
    "AI in Healthcare",
  );

  List<Student> students = [alice, bob];

  print("=== STUDENT INFORMATION ===");
  for (Student student in students) {
    student.displayInfo();
  }

  print("=== PERFORMANCE EVALUATION ===");
  alice.evaluatePerformance();
  bob.evaluatePerformance();
  print("--------------------");

  print("=== ADVISOR REVIEW ===");
  AcademicAdvisor advisor = AcademicAdvisor("Dr. Smith", "Computer
```

```
  Studies");
50   advisor.reviewStudent(alice);
51   advisor.reviewStudent(bob);
52
53   print("=== ASYNCHRONOUS DATABASE SAVE DEMO ===");
54
55   print("Pre-await message: Sending save request for Alice...");
56   String saveResultAlice = await alice.saveRecord();
57   print(saveResultAlice);
58
59   print("\nPre-await message: Sending save request for Bob...");
60   String saveResultBob = await bob.saveRecord();
61   print(saveResultBob);
62
63   print("=== PROGRAM COMPLETED ===");
64 }
65
```

Result:

```
=== STUDENT INFORMATION ===
Name: Alice
Age: 20
Course: Information Technology
Grade: 90.5
Year Level: 3rd Year
Status: Undergraduate
--------------------
Name: Bob
Age: 25
Course: Computer Science
Grade: 85
Thesis Topic: AI in Healthcare
Status: Graduate
--------------------
=== PERFORMANCE EVALUATION ===
Alice is passing their undergraduate modules.
Bob is meeting graduate research standards.
--------------------
=== ADVISOR REVIEW ===
Advisor Dr. Smith (Computer Studies):
Alice Honors Status: Eligible
--------------------
Advisor Dr. Smith (Computer Studies):
```



National Teachers College

629 J Nepomuceno St, Quiapo, Manila, 1001 Metro Manila

```
Bob Honors Status: Not Eligible
--------------------
=== ASYNCHRONOUS DATABASE SAVE DEMO ===
Pre-await message: Sending save request for Alice...
  [System] Initiating database save for Alice...
  [System] Success: Record for Alice saved to database.

Pre-await message: Sending save request for Bob...
  [System] Initiating database save for Bob...
  [System] Success: Record for Bob saved to database.
=== PROGRAM COMPLETED ===
```